



A Novel Algorithm for Identifying and Analyzing Humanitarian Relief Logistics Problems: Studying Uncertainty on the Basis of Interaction with the Decision Maker

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Abstract

Integration of decision-making in relief logistics intends establishing a better condition in fast response. Also, regarding the unpredictability of relief demand, uncertain conditions should be studied in a more applicable way of the planning process. This paper investigates a comprehensive novel algorithm in order to analyze a multi-level location-allocation routing emergency problem in uncertain conditions considering a fast and stable response to the different situations. Due to this reason, system dynamics and its control are studied. In the presented model, the demand is simulated by Mont Carlo simulation. Sum of allocated resources is determined by the expected value of Poisson's distribution based on the approximation of its confidence interval. This model is capable to cover all demands existing in the network. Moreover, allocation of ambulances is considered based on the fuzzy cost. A simulated annealing (SA) algorithm is developed to solve the problem. The obtained results show that the proposed algorithm has good performance and it has been found out that total cost of the emergency system decreases with the increase of covering of stochastic demands, and the responsiveness in each scenario increases. However, the proposed algorithm is highly able to recognize the problem space to create a more precise and adequate description of the problem regarding the conditions.

Keywords Emergency · Ambulance · Location-allocation · Routing · Uncertainty

Introduction

Nowadays, despite growing trend for technological advances, natural disasters (such as earthquakes, floods, hurricanes, avalanches, tornados, fires, volcano eruption, etc.) and unnatural disasters (such as wars, terrorist attacks, road accidents, industrial accidents, political disruptions, refugee crises, etc.) are main restrictions for sustainable developments of countries. The process of management, planning, and control of relief resources for the injured people and patients is called relief logistics. Relief logistics seek to provide the best possible

relief services to the people in need of relief and first aids with the available resources (Overstreet et al. 2011).

Disasters are recognized by their uncertain and unpredictable natures; therefore, demand may change rapidly in such an environment (Swiss 2015). Performance/efficiency of emergency services is determined in terms of the response time and the total cost of logistics (Shafia and Rahmaniani 2013). The road accident rate is 20 times higher than the world average in Iran. According to the Iranian data, one person loses his/her life every 19 min due to the car accidents (Bahadori et al. 2010). These issues made us do some research on the problem.

The primary serious studies on transportation problem during a disaster were done by Knott in 1987. He presented a mathematical model for a situation in which there are a number of different vehicles in each station. His aim was to minimize the total number of uncovered demands (Knott 1987).

Özdamar et al. (2004) applied a research on logistical planning during a disaster. Their aim was to create an effective planning for the distribution of relief aids. They developed a plan for a time-dependent transportation problem. They found

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the optimal solutions for providing relief supports in certain periods of time considering minimum service time.

Doerner et al. (2007) presented a model for minimizing total uncovered demands during a disaster. This objective is equal to maximizing total covered demand. In their model, a central depot was considered to provide services. The aim was to find the best locations for medical services and the best routes for ambulances. Martinez et al. (2011) specify the important role of fleet management improvements in humanitarian logistics. Demand points were considered as a number of small villages with small populations. They solve the model by a heuristic algorithm, and the obtained results were compared with each other. It has been clarified that the usage of OR to adapt supply chain best practices to HL may lead to significant improvements (Van Wassenhove and Pedraza Martinez 2010).

Ortuño et al. (2013) surveyed a literature review on the decision aid models and management systems applied to humanitarian logistics (HL) in this context. Holguín-Veras et al. (2013) proposed the use of social cost as an objective function for post-disaster HL models. Liberatore et al. (2013) analyzed uncertainty in HL based on 27 published papers. Demand, demand location, affected areas, supply, and transportation network are determined as five stochastic parameters considering uncertainty in HL.

In humanitarian relief logistics transportation, location-allocation problems aim to minimize the transportation time regarding the servicing, budget and resource constraints, and emergency demands to increase the lifespan of the injured (Edrissi et al. 2013). However, based on the distribution, humanitarian relief logistics models have different objectives such as considering total cost which includes minimizing traveling costs, distribution costs, manpower costs which their feasible solution is proportional to the extent of supply (Liberatore et al. 2014). On the other hand, the capacity of facilities is one of the main elements in these problems regarding the demand covering. Therefore, issues such as inventory amount, resource flow, and the number of injuries belong to this category.

Chou et al. (2014) applied the biological-based GA for allocating disaster refuge site staff and planning relief supply distribution. They compared the proposed algorithm with immune algorithms and GAs in terms of effectiveness. Salman and Gül (2014) proposed a multi-period MIP for dynamic casualty transportation model for determining the location and size of emergency service facilities to be established after a disaster to cope with the demand surge.

Walter and Gutjahr (2014) developed a location-routing model to provide relief supports for affected people during a disaster. They claimed that intermediate emergency stations shall be established temporarily in order to respond to the requirements of people in the possible shortest time. They proposed a three-objective model for minimizing short-term

and mid-term costs. They used the ϵ -constraint method to solve their model.

Abounacer et al. (2014) propose a model to determine the optimal number of relief centers and humanitarian aid distributions. They also tried to determine the best routes for delivering aids to demand points. They considered three different objectives. The first objective was to minimize the time of delivering aids to the affected areas. The second objective was to minimize the number of established centers, and the third one was to minimize total uncovered demands. They found the optimal solutions for the problem using the ϵ -constraint method and by balancing the objectives.

Rath et al. (2015) proposed epsilon constraint method for maximizing demand coverage and minimizing the total cost objectives. Huang et al. (2015) formulated three-objective functions for HL as a lifesaving utility, delay cost, and fairness. They designed efficient variation inequality algorithm for MO optimization model that combines resource allocation with emergency distribution.

Knyazkov et al. (2015) investigated the dynamic routes of ambulances for transferring injured people in Saint Petersburg. The objective was to shorten the time of transferring injured people to the hospitals. This was done by determining an optimal route to the nearest hospital in the city. This made the problem very complicated. They included a number of features of urban areas in their study related to the urban areas, such as traffic, population congestion, the capacity of hospitals, etc.

Chen and Yu (2016) studied a location-routing problem for offering medical emergency services during a disaster to the demand points. The objective was to increase the efficiency of services during a disaster. They presented a mixed integer linear programming and a relevant graph network in order to find the optimal locations for facilities. They used Lagrange relaxation method to solve the problem. Finally, they have done a case study and analyzed the obtained results to evaluate their model applicability. Babaei and Shahanaghi (2017a, b, c) write three papers on areas of emergency medical services considering to municipal issues.

After reviewing the important publications in literature, it is clarified that just few papers have tackled the integrated problems together, i.e., there is lack of contributions belong to the integration of real world conditions and possibilities such as uncertainty, facility location, routing, and simulation in estimating demand value. Also, the gaps in existing methodologies are noticeable in integrating the procedure of defining, analyzing, simulating, developing models, and solution methods.

Existence of the limited financial source for locating and managing of the ambulance stations to provide services is the main reason why planning is highly necessary for this field. In addition, because of uncertain nature of demands in preparing mobilized ambulances and the uncertain traversing time of

them through the routes in the different time periods of the day, uncertain planning will yield more real applications.

Generally, a significant missing from the aforementioned research is not considering the stability of the problem space initially. By omitting this gap, the decisions will be made by countable and robust results. On the other hand, considering the integration of the model in determining the optimal location for the emergency stations, optimal allocation of the ambulance, and optimal routing can lead to obtaining better and more precise results. According to these points, a proposed novel comprehensive algorithm is studied for implementing allocation and routing of ambulance, location of emergency centers under uncertain condition through (1) identifying problem space, (2) investigating the stability of the problem, (3) categorization of demand parameter, (4) determining the probability of different scenarios, (5) prioritizing the possible choices using TOPSIS method and LINMAP model, (6) developing a respective mathematical model (a hybrid robust multi-step planning is used which relies on chance programming due to the interaction with the decision makers), (7) developing a solution method (a SA algorithm), and (8) presenting the obtained results and related analysis. In the following, in Section 2, the proposed model is presented. In Section 3, numerical analysis is provided to analyze the model. Finally, in Section 4, the concluding remarks are presented and suggestions for further studies are recommended.

Problem Definition

The planning for the ambulances in relief logistics is one of the most important issues which encounters with several challenges that make the decision-making more complex. The proposed solution algorithm using in this paper is represented in Fig. 1.

The steps of the algorithm are explained in the following subsections.

System Dynamics and Control

System dynamics is one of the primary responses to the infirmity of the operational research and the other techniques of the management science which were presented by Forrester (1961) at MIT University for the first time.

To determine the space of the emergency problem which is going to be modeled, a causal diagram of system dynamics can be used. By using the causal diagram, the flow diagram of the emergency problem can be determined by recognizing the variables and the levels. The diagram in Fig. 2 is the flow diagram of the emergency problem which has been described.

Structure of a complex feedback loop system is not simple and in which a state of the system specifies how to behave. A complex system has many feedback loops. The rates and internal levels of it are non-linear. The complex system consists of a high order which means that the number of the level variables is high. These systems are usually involved with positive loops which show the growth processes. Also, it has negative or purposeful loops. As it can be seen in Fig. 2, the problem includes two rates. The input rate which is proportional to the rate of service coverage and the cost rate which is proportional to the rate of emergency demand or in other words, the level and the number of the patients/injured. This means that it shows the level of unsatisfied demand. The rate of service covering is rarely equal to the demand in the real world, and as a result, the capacity of the responsiveness would be specified as a percentage of the demand. The responsiveness capacity is also proportional to the intersection of the number of the untreated patients, the minimum capacity of the emergency station, and the reception capacity of the hospital. As the constraints generate the feasible space of decision-making, the minimum capacity of the hospital and the emergency station is regarded as the basic capacity.

$$c' = \text{Min}(A, B) \quad (1)$$

As it can be seen in Fig. 3, the block diagram is presented based on the causal diagram. Also, the signal diagram is

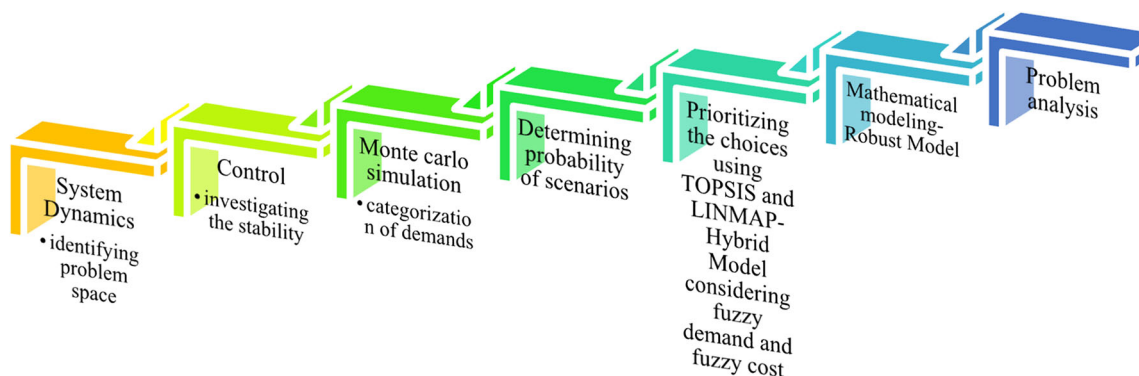
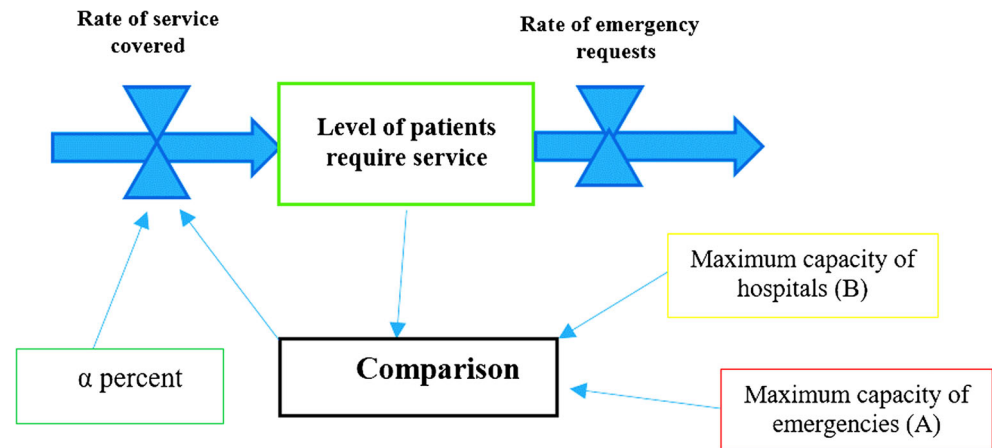


Fig. 1 The proposed solution algorithm

Fig. 2 Proposed flow diagram



shown in Fig. 4. These diagrams are adapted from Ogata's book in proportion to our problem (Ogata 2010). Details of applied symbols are presented in Table 1.

In order to make a decision on such problems and, determine and improve the problem space, it is critical to ensure that the decision space is stable, because making decisions in unknown and highly variable circumstances do not lead to appropriate results. Thus, decision space must be stable in order to consider uncertainty in the problem and make the problem closer to the real world. The control step is presented according to Ogata's book (Ogata 2010) and it can be proved that problem is controllable.

To investigate the stability of a system, transformation function of the system must be determined. Then, according to the flow equations, blocked and signal equations (which were represented in the previous section), the Mason interest can be calculated (see Appendix 1).

Simulation

In this paper, demand is defined by intensity and probability value. In this way, demand is introduced by six scenarios based on its probability. The considered probability value and the demand intensity are presented in Table 2 (Yuan and Puchalsky 2015).

The cumulative diagram of Table 1 can be presented using Monte Carlo simulation. Also, using random numbers corresponding to the scenarios, uncertainty condition

is transformed into the deterministic one. The results are represented in Table 3.

In Table 4, demand is represented with the probability of the event.

Also, by using the Monte Carlo simulation, final results are given in Table 5.

By doing so, demands are transformed into a deterministic (crisp) value.

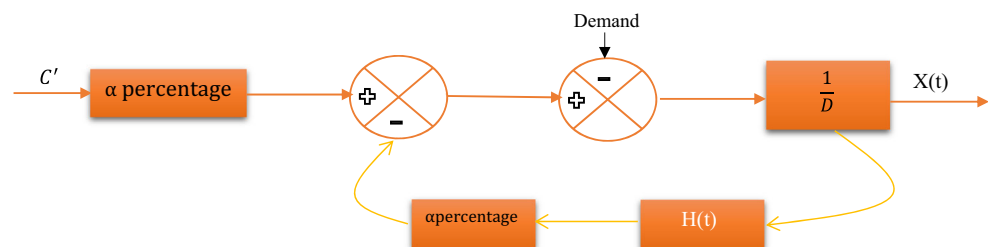
For instance, consider a city which needs to have relief logistics. A city can be divided into three sections and demand for each section needs to be determined through the above-mentioned approach. While the demand for a section is dependent on other sections, it can be a coefficient of the specified section with respect to the relation type. For example, if the eastern area of the city is assumed as a host of employee's traffic in the morning, the other areas of this city will be host at the end of the work shift. Thus, the incidents are higher in the morning for the eastern area and it is higher in other areas at the end of the work shifts.

Determining the Probability of the Scenario Occurrence

Scenarios have been extracted from the model below based on two criteria due to the utilities (Wang et al. 2016; Liu and Liu 2009). A small example has been considered and analyzed by the proposed model of Wang et al. (see Appendix 2).

The results of solving the model are shown in Table 6.

Fig. 3 The block diagram



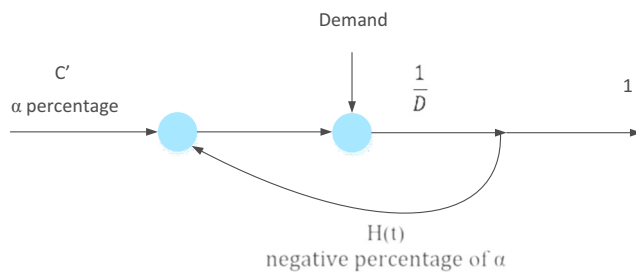


Fig. 4 The signal diagram

Prioritizing the Choices

The locations which are chosen to establish emergency stations should be investigated by three criteria. Criteria such as efficiency, compatibility, and safety can prioritize the intensity and stability in these candidate locations. Due to this purpose, efficiency can be considered as accessibility of the emergency location to the main routes, urban areas, and its domain. On the other hand, compatibility can be considered in a geological viewpoint and safety is in turn considered from the perspective of privacy observance by incompatible applications with the emergency location. Also, in the efficiency criterion, accessibility can be proportional to the variable and uncertain traffic. Thus, four locations which are specified by experts are represented in Table 7.

The experts' opinions for the uncertain efficiency are on the basis of a normal distribution with the given average, variance, and confidence percentage. In deterministic efficiency, this uncertain criterion has been changed to a deterministic value and the last three columns are the basis for decision-making. By Topsis method, the importance coefficient of each location becomes as what has been shown in Table 8.

Therefore, the last column is used as cost coefficient in the mathematical model (Araghi et al. 2011; Opricovic and Tzeng 2004).

Occasionally, after determining the priority of choices, decision maker expects that priorities be different from the mathematical method. This difference can originate from the previous experiences and expertise of the expert. To interact with the decision maker, his opinions should be supported toward the priorities and as a result, LINMAP method can be used to test the choices which can be replaced (Hajiagha et al. 2012).

Table 1 Details of proposed diagrams

Symbol	Mathematical base	Description
c	$\text{Min}(A, B)$	Minimum capacity of hospital
Demand		Demand of injured people
$X(t)$		System feedback in t
α		Coefficient as a percentage
$H(t)$		System feedback in t
$\frac{1}{D}$	Integral based on dt	Update uncovered demands

Table 2 The probability value and the demand intensity

Probability	0.05	0.15	0.3	0.25	0.15	0.1	$\sum p_i = 1$
Demand intensity	5	4	3	2	1	0	

The linear programming technique for multidimensional analysis of preferences is known as LINMAP which is one of the well-known seeking methods for multi attribute decision-making problems (Hajiagha et al. 2012). This method is basically proposed under crisp and deterministic circumstances.

In this paper, a LINMAP model has been presented by using fuzzy and stochastic concepts such that the difference between the desirable choice of the mathematical model and decision maker is stochastic. Anyway, the proposed LINMAP model is represented as follows.

LINMAP Model

$$\begin{aligned}
 & \text{Max } Z \\
 & \text{St.} \\
 & Aa_i y_i + Z(1) \leq A^- a_i \quad \forall_i \\
 & y_i \geq 0 \quad \forall_i \\
 & Z : \text{Free Variable}
 \end{aligned} \tag{2}$$

In this model, z is freedom variable and if it becomes zero, the capability of changing the priorities become possible. $A\alpha_i$ is proportional to transpose of paired difference matrix of the choices relative to the criteria. y_i is the representative of the difference between each pair of the choices and if it takes positive values, it has the capability of changing the priority of the choices. $A^- \alpha_i$ indicates paired difference matrix of the choices which their priority is being investigated to see whether a change can be made in them.

LINMAP-Hybrid Model

$$\begin{aligned}
 & \text{Max } Z \\
 & \text{St.} \\
 & \varnothing_{(\theta)}^{-1} A\alpha_i y_i + Z(1) \leq A^- \alpha_i - \overline{A}^- \alpha_i \quad \forall_i \\
 & A\alpha_i y_i + Z(1) \leq (A^- \alpha_i - d)(1-2\theta) + 2\theta \overline{A}^- \alpha_i \quad \forall_i \\
 & y_i \geq 0 \quad \forall_i \\
 & Z : \text{Free Variable}
 \end{aligned} \tag{3}$$

Table 3 The certain intensity of demand

Random generated number	0.23	0.58	0.57	0.78	0.26
Intensity of certain demand	2	3	3	3	1

Table 4 The probability and demand values

Probability	0.1	0.15	0.25	0.2	0.1	0.1	0.05	0.05
Demand	0.4	35	30	25	20	0.15	10	5

In the presented model, $\varnothing_{(\theta)}^{-1}$ shows the probability of estimating the differences of paired choices from difference matrix of paired choices and it is calculated by subtracting it from column average of the matrix ($\overline{A} \alpha_i$), that is calculated positively. Constraint (24) b, which is adjusted for the confidence percentage up to 50, estimates the percentage of the pair difference which should be replaced. Note that this percentage is proportional to θ . In this constraint, d represents the lower bound of the fuzzy difference. The model is solved with constraint (24) and then with constraint (24) b. So, the minimum of the intended solution would be the final solution which is applicable in both situations. For instance, if in the prioritizing table of two decision makers, choice of two would be better than one therefore, with regard to the estimated probability for 20% confidence and the fuzzy limit of 5 and the average of each column, the solution is accepted. While in other problems, the problem is incapable of satisfying the condition or the feasible space is not applicable. Thus, stochastic satisfaction is able to make the experts' opinion possible.

If in the model, z be zero, then the replacement of the priorities is done proportionally to the decision maker's opinion. But, if its value is not zero, there is no possibility to replace. Anyway, to interact with the decision maker more, it should be clarified that how to accommodate his expectation is deviated with the given values. So, for responsiveness that results in getting closer to the opinions and expectations of the decision maker, the model below is proposed.

$$\begin{aligned} \text{Obj1 : Min } w_p |x - A_p| + w_l |x - A_l| & \quad (a) \\ \text{Obj2 : Max } w_p |x - A_p| + w_l |x - A_l| & \quad (b) \\ \text{St. } x \geq 0 & \end{aligned} \quad (4)$$

The model has two objectives. x shows the value that can be replaced with the desirable choice. A_p is the value of the desirable choice with the mathematical method in a specific criterion and A_l is the desirable expected choice of decision maker in the same criteria. In this model, w_p and w_l are the weights which show the importance degree of objective estimation than desirable choice obtained by the mathematical method or the expected method. These weights are different in objectives (25) a and (25) b. Here, the closeness and remoteness of a choice are shown in objectives (25) a and (24) d,

Table 5 Confirmed values

Random number	0.21	0.4	0.15	0.74	0.1	0.6	0.69	0.4	0.28	0.74	0.75	0.85
Intensity	2		3			3			3			1
Demand	65		75			70			50			20

Table 6 The obtained values of the model

Values obtained from the model	p
0.6	1
0.4	2
0.4	3
0.6	4
0.4	5

respectively. To reach a better performance, the model is linearized as below:

$$\begin{aligned} \text{Min } d \\ \text{St.} \\ -\theta_p + A_p - x \leq 0 \\ -\theta_l + A_l - x \leq 0 \\ w_p (x - A_p + 2\theta_p) + w_l (x - A_l + 2\theta_l) = Z_{Lobj1} \\ w_p (x - A_p + 2\theta_p) + w_l (x - A_l + 2\theta_l) = Z_{Lobj2} \\ Z_{Lobj1} - Rd \leq \alpha_p \quad (a) \\ Z_{Lobj1} + Qd \geq \alpha_p \quad (b) \\ x, \theta_l, \theta_p \geq 0 \end{aligned} \quad (5)$$

In this model, θ_p and θ_l are the linearization variables, R and Q are the weights for the deviation amount of closeness or remoteness to or from the choices. α_p is the ideal value of solution closeness to the expectations of the decision maker which can be different in constraints e and f. As far as this value is closer to the calculated value of the mathematical method, the difference with the desirable choice of the mathematical method gets nearer to zero, and it would be better. By the value of d , it can be perceived that decision maker should accept how much of the solution deviation to reach to its desirable choice. The basis of this method is goal attainment and minimization of the variances.

Presenting the Model

Considering integrated planning in relief supply chain constructs the basis of this research, which includes determining the location of the stations, allocating ambulances to them and also ambulances routing to relieve the patients in a standard time and with the minimum cost.

First, consider a city that requires being relieved with first aids when a disaster occurs. Problem space can be depicted as a graph network; therefore, some nodes of the graph (in the considered city) are the candidates to be established as an emergency station for the ambulances and also, stations can be established with

Table 7 Experts' opinion on the candidate locations

Safety	Compatibility	Deterministic efficiency	Confidence percentage	Uncertain efficiency	
3	400	56.45	0.9	(50.25)	Location one
5	450	49.95	0.95	(45.9)	Location two
9	700	78.97	0.99	(58.81)	Location three
7	470	50.28	0.85	(43.49)	Location four

different costs on these nodes. These centers are the depot of the ambulances which they begin their tour from there in order to go to the service-required nodes and then, to the hospitals. So, different paths can be chosen. The optimal routes of the ambulances are in conjunction with optimal objective value.

Traffic load on these paths depends on the time of traversing and because of this, a single approach cannot be chosen for all the times. Consequently, different scenarios should be simulated. This traffic load has a direct impact on the traversing time. It is possible that a long time would be necessary for a short path and vice versa. Therefore, it is more appropriate to consider the time instead of the path between two nodes. Finally, it should be decided about the selection of the destination hospitals. The possibility of transferring the ambulances is also considered in order to have a better use of available resources. On the other hand, due to the uncertainty of demand for the ambulances in the stations, demands are considered to have scenario-based values.

The proposed model has multiple steps, that in one step, the assignment is based on the scenarios for covering the demands. In the next step, the final assignment for mobilizing the ambulances in the stations based on the assigned scenario is conducted and then, based on the final assignment, the locations of the required emergencies are established. Finally, after determining the location of the emergency stations, the routes are constructed proportional to them. Therefore, the proposed model is a multi-step one. In next section, the model is presented.

Consider a city with seven applicants, one hospital and two potential emergency stations that need to be integrally planned and handled after a disaster occurred. The constructed routes are shown using arrows toward the demand nodes. Only one of the two potential emergency stations is established with respect to the total demand value. This example has been depicted in Fig. 5.

Table 8 Prioritizing the locations

Cost priority	Main priority	Location
0.94	0.06	1
0.76	0.24	2
0.13	0.87	3
0.55	0.45	4

After simulating the demand and determining the priority choices according to LINMAP, the proposed models can be presented.

Indexes

- I Applicants
- J Emergency stations
- K Hospitals
- S Scenarios

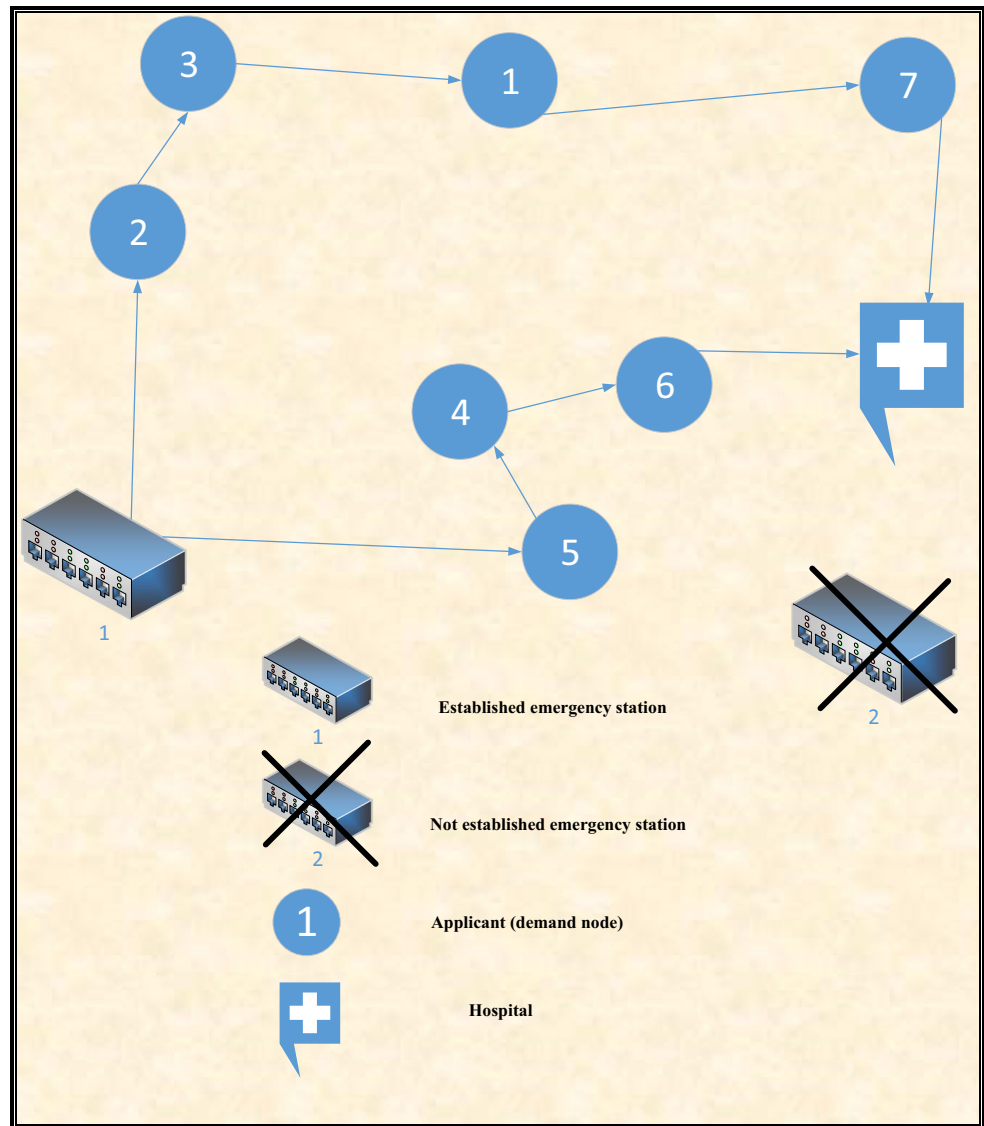
Parameters

- d_{is} The quantity of demand i in scenario S
- T The standard time
- α : Minimum cover percentage
- t_{ijk} The traversing time from emergency station j to the demand i and from then, to the hospital k
- M A large number
- f_j Establishing cost for emergency location j
- g_j Cost of assigning the ambulance to the emergency station j
- C_s Cost of required ambulance for demand i from the station j under the scenario S
- A Cost of establishing a path from i to j up to k
- B Cost of establishing a special path from i to j and to k
- Bb budget
- b_s Performance coefficient for assignment in each scenario
- ω Coefficient of feasible space
- λ Coefficient of scenario effects
- dd The difference between fuzzy central number with the lower bound
- θ Confidence percentage
- $\bar{d}_{is}$ The average of demands
- $\tilde{C}$ The cost of fuzzy assignment
- $E(\tilde{C})$ Expected value of $\tilde{C}$

Decision-making Variables

- X_j One, if location j is established as the emergency station, otherwise 0.

Fig. 5 The schematic example



- y_{ijs} The number of needed ambulance for demand i in station j under scenario S
- z_j The number of assigned ambulances to the emergency station j
- x_{ijk} One if the ambulance goes from emergency j to the demand i and then to the hospital k , otherwise 0.
- x_{ijk}^E One if the ambulance goes from emergency j to the demand i and then to the hospital k through a special path, otherwise 0.
- V_{ijs} V : the transformer variable

Robust Model

Under certain conditions, the robust approach does not increase the complexity degree of the problem significantly (Tirkolaee et al. 2017). For improving the model in order to apply the final assignment of the ambulance in accordance

with the number of required ambulances for each demand point, the deviation from the feasible space is considered. Also, the impact of other scenarios on each scenario is considered as an ameliorative factor of the model and the objective function. This means that in each scenario, the number of the required ambulance in each demand point causes the differences between the other scenarios.

$$\begin{aligned} \text{Min } \sum_j (f_j, x_j + g_j, z_j) + \sum_i \sum_j \sum_k (AX_{ijk} + BX'_{ijk}) + \sum_s \sum_i \sum_j p_s y_{ijs} \quad (6) \\ + \lambda \sum_s p_s \left(\sum_i \sum_j y_{ijs} - \sum_{s'} p_{s'} \sum_i \sum_j y_{ijs} + 2\theta_s \right) + w \sum_s p_s (\delta_s + 2\Delta_s) \end{aligned}$$

$$-Mx_j + z_j \leq 0 \quad \forall_j \quad (7)$$

$$\sum_i y_{ijs} - b_s z_j + \delta_s = 0 \quad \forall_{j,s} \quad (8)$$

$$\alpha \sum_i \sum_s d_{is} - \sum_i \sum_j \sum_s y_{ijs} \leq 0 \quad (9)$$

$$\sum_j y_{ijs} - d_{is} \leq 0 \quad \forall_{i,s} \quad (10)$$

$$t_{ijk} x_{ijk} - MX'_{ijk} - T \leq 0 \quad \forall_{i,j,k} \quad (11)$$

$$x_j - \sum_i \sum_k x_{ijk} \leq 0 \quad \forall_j \quad (12)$$

$$y_{ijs} - M \sum_k x_{ijk} \leq 0 \quad \forall_{i,j,s} \quad (13)$$

$$-\theta_s - \sum_i \sum_j y_{ijs} + \sum_s p_s \sum_i \sum_j y_{ijs} \leq 0 \quad \forall_s \quad (14)$$

$$-\Delta_s - \delta_s \leq 0 \quad \forall_s \quad (15)$$

$$\sum_i \sum_k x_{ijk} \leq M z_j \forall_j \quad (16)$$

$$\begin{aligned} y_{ijs} \geq 0, Z_j \geq 0 \quad \text{int}, \forall_{i,j,s} \\ x_{ijk}, X_j, x'_{ijk} \in \{0, 1\} \forall_j \\ \Delta_s, \theta_s \geq 0 \end{aligned} \quad (17)$$

Objective function (7) includes five parts. The first part minimizes total cost of establishing the emergency location and the cost of ambulance assignments to the emergency station. The second part is to minimize the cost of constructing the path and the special path. The third part aims to minimize the number of the required ambulances in each scenario. The fourth part represents the impact of other scenarios on the other ones. The fifth part investigates the feasibility of the problem.

When the assignment of ambulances to the stations is done, a location must be established in that place as a station (Eq. (7)).

The final assignment of a location is directly conjunct with the number of ambulances that is demanded in each scenario (Eq. (8)).

The total number of ambulances that are required in each scenario should be at least α percent of total demand (Eq. (9)).

The number of the required ambulances for each demand in each scenario would be less than the demand (Eq. (10)).

The total time needed to construct a path must be less than the standard time T . Otherwise, the path will be transformed to a special path (Eq. (11)). In simple words, when total time for the process of moving from the emergency station up to carrying injured people to the hospital is more than the standard time (which is calculated by the experts based on the probability of injured survival) then, a special path of emergency is provided which reduces the time in terms of management and physical structure. In management terms, it is based on the traffic control of the region and in structural terms, it can help by establishing an emergency path.

The path is constructed when an emergency station is established (Eq. (12)). The path is constructed according to the number of assigned ambulances (Eq. (13)). Equations (14)

and (15) are the limitations of robust linearization. Equation (16) guarantees the formation of a path of the ambulance station. Type of variables is specified in constraints (17).

Mathematical Model for Allocation of Uncertain Servicing Resources

In this section, the required value for each demand point of emergency location is considered stochastically. Also, the resources of the supplier are coped with uncertainty in covering the requirements. As we can say, each demand point is to maximize the probability of satisfying its own emergency requirements due to the fact that the resources which employ are coped with uncertainty.

$$\text{Max } \bar{f} \quad (18)$$

$$-Mx_j + z_j \leq 0 \quad \forall_j \quad (19)$$

$$\sum_i y_{ijs} - Z_j \leq 0 \quad \forall_{j,s} \quad (20)$$

$$\alpha \sum_i \sum_s d_{is} - \sum_i \sum_j \sum_s y_{ijs} \leq 0 \quad (21)$$

$$\sum_j y_{ijs} - d_{is} \leq 0 \quad \forall_{i,s} \quad (22)$$

$$t_{ijk} x_{ijk} - MX'_{ijk} - T \leq 0 \quad \forall_{i,j,k} \quad (23)$$

$$x_j - \sum_i \sum_k x_{ijk} \leq 0 \quad \forall_j \quad (24)$$

$$y_{ijs} - M \sum_k x_{ijk} \leq 0 \quad \forall_{i,j,s} \quad (25)$$

$$\sum_j (f_j x_j + g_j z_j) + \sum_i \sum_j \sum_k (A x_{ijk} + B x'_{ijk}) + \sum_i \sum_j \sum_s y_{ijs} \leq Bb \quad (26)$$

$$\sum_j y_{ijs} \leq \mu \bar{f} \forall_{i,s} \quad (27)$$

$$\sum_i \sum_k x_{ijk} \leq M z_j \forall_j \quad (28)$$

$$\bar{x} - Z_{\frac{\gamma}{2}} \sqrt{\frac{\bar{x}}{n}} \leq \mu \leq \bar{x} + Z_{\frac{\gamma}{2}} \sqrt{\frac{\bar{x}}{n}} \quad (29)$$

$$\begin{aligned} y_{ijs} \geq 0, Z_j \geq 0 \quad \text{int}, \forall_{i,j,s} \\ x_{ijk}, X_j, x'_{ijk} \in \{0, 1\} \forall_j \end{aligned}$$

$$\bar{f} \text{ free variable} \quad (30)$$

The objective function is to increase the probability of the ambulance allocation to the locations that satisfies the demand in each scenario (Eq. (18)).

Equations (19) to (25) and (28) are repeated.

Equation (26) shows that the financial resources (budget) are always a fixed sum.

Equation (47) shows that total required ambulances follow the average of Poisson's distribution in each scenario for each demand point. The impetus behind using this distribution is that the occurrence probability of the number of these events is related to the time duration and also, number of events in separated time intervals is independent of each other.

Equation (29) is the confidence distance for μ in Poisson's distribution. In this constraint, $\bar{x}$ is the average, n is the sample size and $\bar{x} \pm Z_{\frac{\gamma}{2}}\sqrt{\frac{\bar{x}}{n}}$ is the one-sided confidence level with the confidence level of $(1 - \gamma)\%$. Type of the variables is specified in constraint (31).

Robust Model with Uncertainty Servicing Resources and Considering the Probability of Satisfying the Fuzzy Demand with Fuzzy Costs

In this model, total cost of assigning each point to the ambulances is proportional to the location of emergency stations and it is a fuzzy triangular. Therefore, the model aims to increase the responsive chance for the needs of each demand point. In addition, the assigned cost of each demand point to the ambulance is related to the emergency stations and it is a fuzzy triangular number. Hence, the model is to increase the chance of responsiveness to the needs of each demand point.

$$\text{Max } \bar{f} - \sum_i \sum_j \sum_s p_s y_{ijs} - \lambda \sum_s p_s \left(\sum_i \sum_j y_{ijs} - \sum_s p_s \sum_i \sum_j y_{ijs} + 2\theta s \right) - w \sum_s p_s (\delta^2 + 2\Delta s) - \sum_s \sum_i \sum_j V_{ijs} \quad (31)$$

$$\text{S.t.} \\ -Mx_j + z_j \leq 0 \quad \forall_j \quad (32)$$

$$\sum_i y_{ijs} - b_s z_j + \delta_s = 0 \quad \forall_{j,s} \quad (33)$$

$$\alpha \sum_i \sum_s d_{is} - \sum_i \sum_j \sum_s y_{ijs} \leq 0 \quad (34)$$

$$\sum_j y_{ijs} - d_{is} \leq 0 \quad \forall_{i,s} \quad (35)$$

$$\varnothing_{(\theta)}^{-1} y_{ijs} \geq d_{is} - \bar{d}_{is} \forall_{i,j,s} \quad (36)$$

$$y_{ijs} \leq (1-2\theta)(d_{is} - dd) + 2\theta d_{is} \forall_{i,j,s} \quad (37)$$

$$t_{ijk} x_{ijk} - Mx'_{ijk} - T \leq 0 \quad \forall_{i,j,k} \quad (38)$$

$$x_j - \sum_i \sum_k x_{ijk} \leq 0 \quad \forall_j \quad (39)$$

$$y_{ijs} - M \sum_k x_{ijk} \leq 0 \quad \forall_{i,j,s} \quad (40)$$

$$-\theta_s - \sum_i \sum_j y_{ijs} + \sum_s p_s \sum_i \sum_j y_{ijs} \leq 0 \quad \forall_s \quad (41)$$

$$-\Delta_s - \delta_s \leq 0 \quad \forall_s \quad (42)$$

$$\sum_i \sum_j \sum_s V_{ijs} = V \quad (43)$$

$$\sum_i \sum_j \sum_s E(\tilde{c}) p_s y_{ijs} \leq V \quad (44)$$

$$\sum_j (f_j x_j + g_j z_j) + \sum_i \sum_j \sum_k (Ax_{ijk} + Bx'_{ijk}) + \sum_i \sum_j \sum_s y_{ijs} \leq Bb \quad (45)$$

$$\sum_j y_{ijs} \leq \mu \bar{f} \forall_i, \forall_s \quad (46)$$

$$\sum_i \sum_k x_{ijk} \leq Mz_j, \quad \forall_j \quad (47)$$

$$\bar{x} - Z_{\frac{\gamma}{2}} \sqrt{\frac{\bar{x}}{n}} \leq \mu \leq \bar{x} + Z_{\frac{\gamma}{2}} \sqrt{\frac{\bar{x}}{n}} \quad (48)$$

$$\bar{f} \text{ free variable, } V_{ijs}, V \geq 0 \\ Z_j \geq 0, X_j \in \{0, 1\}, x_{ijk}, x'_{ijk} \in \{0, 1\}, \Delta_s, \theta_s \geq 0, V_{ijs}, V \geq 0 \quad Qu_{is} \geq 0 \quad (49)$$

The objective function has five parts (Eq. 31). The first part is to increase the probability of allocation of the ambulance to the locations covering the demand in each scenario. The second part is to minimize the number of required ambulances in each scenario. The third and fourth parts are adopted from main components of the robust model. The third part shows the effect of other scenarios in the current scenario. The fourth part investigates the feasibility of the problem. The fifth part focuses on the minimization of the number of ambulances in order to minimize the costs.

The number of required ambulances for each demand in each scenario should be less than the demand (Eq. (35)).

For demand, the chance of being a proportion of the demand point in each scenario from each location is based on the distribution probability. Here, the distribution of the probability coefficients of the variable is based on the normal distribution with the average of zero and variance of one (Eq. (36)).

The chance of generating a ratio of the demand point in each scenario from each location is based on the fuzzy defined demand. For this constraint, it is assumed that the maximum confidence is 50% (Eq. (37)).

To solve the problem, the minimum of the Eq. (36) and Eq. (37) are considered. The solution in both of the problems is feasible but the better problem can be selected according to its conditions.

Solution Method

In this section, a heuristic and a metaheuristic algorithm (namely Simulated Annealing (SA)) is developed to solve

Emergency station				
0	0	0	1	0

Fig. 6 Solution representation for initial solution

the proposed robust model through generating initial solutions and improving them. The heuristic is applied to generate initial solutions; the SA is applied to improve these solutions. This makes an appropriate efficiency in order to generate good solutions within a reasonable run time (Tirkolae et al. 2016).

Solution Representation

In order to generate the binary variables easily observable, a single-row matrix is considered that is related to the locations of the emergency station. The dimension is $1 \times |I|$ in which columns of the matrix are as potential locations for the establishment of the emergency station. This matrix's elements are binary in a way that zero and one correspond to the establishment and the non-establishment of the emergency station.

$$X_j = [X_1 \quad X_2 \quad \dots \quad X_j] \quad (50)$$

For example, the solution representation for this matrix can be written as Fig. 6 if four candidate locations exist for establishing the transfer points.

Then, a constructive heuristic algorithm is developed to generate several initial routes for the ambulances to create services. The steps of the algorithm are described in the following:

- 1- Assign ambulances to the established stations. Assign a random demand point to the ambulance.
- 2- Choose the next demand point with respect to the related constraints such as considering the standard time to service. In addition, demand point which has the least cost of establishing a path and least traversing time should be chosen. If we have multiple points to satisfy these constraints, select one of these randomly and service it.
- 3- If standard time limitation has been exceeded, send another ambulance. Otherwise, service other demand points.
- 4- If all demand points are being serviced, go to 5. Otherwise, go to 2.
- 5- Stop the algorithm. Report the obtained solution.

The solution representation including the constructed routes for the ambulances is presented in Fig. 7. There are some ambulances assigned to the established stations. Each

Fig. 7 Solution representation for initial solution

#Ambulance in established station	Demand point							hospital
1	7	5	2	4	0	0	0	1
2	1	6	3	0	0	0	0	1

Input: Cooling schedule.

$s=s_0$; /* Generation of the initial solution */

$T=T_{max}$; /* Starting temperature */

Repeat

Repeat /* At a fixed temperature */

Generate a random neighbour s' ;

$\Delta E = f(s') - f(s)$;

If $\Delta E \leq 0$ Then $s=s'$ /* Accept the neighbour solution */

Else Accept s' with a probability $e^{-\frac{\Delta f}{T}}$;

Until Equilibrium condition

/* e.g. a given number of iterations executed at each temperature T */

$T=g(T)$; /* Temperature update */

Until Stopping criteria satisfied /* e.g. $T < T_{min}$ */

Output: Best solution found.

Fig. 8 Simulated annealing pseudo code

row in the solution shows the constructed tour of the ambulances. If there is no demand point to service, the related cell is zero.

In our generated example, we consider a network with seven demand points and three potential emergency stations.

Simulated Annealing

Simulated annealing is a technique which has been applied to problems that are both difficult and important. The simulated annealing begins its search from a random initial solution. The iteration loop that characterizes the main procedure randomly generates in each iteration only one neighbor s' of the current solution s . The discord Δ for the value of the objective function (f) is tested for each neighbor generation (Egglese 1990). To test this discord, $\Delta = (f(s') - f(s))$ is calculated. If the value of Δ is less than zero, then the new solution s' will be automatically accepted to replace s . Otherwise, accepting the new solution s' is dependent on the probability established by the Metropolis criteria, which is given by $(-\Delta/T)$, where T is a temperature parameter, a key variable for the method. Therefore, we have the following:

$$P(\text{Acceptance}) = e^{-\frac{\Delta f}{T}} \quad (51)$$

Figure 8 shows the simulated annealing pseudo code.

Local searches used in this algorithm are as below:

1. Change the established station to non-established station and choose another one to be established.

Table 9 Data for sample problem

Parameters	Small-sized instance problem	Large-sized instance problem
Number of applicants	3	15
Number of potential emergency stations	4	8
Number of hospital	3	5
Number of scenario	3	5
T	100	100
ω	0.7	0.7
λ	0.3	0.3
$\varphi_{(\theta)}^{-1}$	1.64	1.64
$\overline{d}_{is}$	90	90
θ	0.2	0.2
$\bar{x}$	600	600
n	100	100
dd	10	10

In constructed routes by ambulances:

2. Reverse the sequence of the serviced demand point in each solution.
3. Select a part of a solution that is selected randomly and changes its sequence.

It is noticeable that feasibility of the new generation of solutions is checked after applying these local searches.

Algorithm Stop Criterion

A maximum number of iterations are considered in order to stop the algorithm. In addition, another stopping criterion is considered as the maximum number of iterations with no improvement in increasing the efficiency and decreasing waste time. Finally, report the solution with best objective value.

4. Results Analysis

Due to the fact that hessian matrix is the limitations of the feasible area and the positive semi-definite objective function and also the problem is linear, therefore, the feasible area of the problem and the objective function are convex. Hence, in this convex planning type, the problem involves the optimum

point which is attained by solving the problem. In the other words, the solution is convergent to the optimum point.

After demand simulation and determination of priority choices according to LINMAP, the problem can be analyzed. The model is solved and analyzed by considering four candidate location for the emergency stations to cover three demand points and to transfer the injured to three hospitals. The problem is with a fuzzy demand under different scenario assignment. Definitions of sets and parameters for the instance problem are shown in Table 9 in details. Also, the parameters are generated randomly.

In the first step, we generate two instances in a different size to evaluate the performance of the proposed algorithm. Then these two instances are solved both by the proposed algorithm and by GAMS software. Detailed information of the instance problems is shown in Table 9. The model is analyzed through the GAMS software by Baron Solver in 1000 s. The obtained result is presented in Table 10.

As it can be seen in Table 10, SA has shown appropriate performance in comparison with GAMS software. Also, large-sized instances solved by SA have better results in comparison with GAMS. In the following, a sensitivity analysis is applied on the small-sized instances to investigate the effect of uncertainty on the other parameters.

The limitation of assignment scenario is based on the probability satisfaction of the demands and being studied in Problem 1 with, and problem 2 is an assignment scenario based on the fuzzy demand. Parameters of ω , λ , $\varphi_{(\theta)}^{-1}$, $\overline{d}_{is}$ are changed in the interval of 20% decrease and 20% increase in order to investigate the sensitivity of the problem to them. Therefore, in tables and figures below, their effect on the objective functions is investigated. The numerical results are shown in Table 11.

As it can be seen in Fig. 9, by increasing parameter ω , total profit decreases. This change interval leads to profit decreasing with an almost constant slope and the relation between this parameter and total profit is approximately linear. In other words, by increasing the assignment of the ambulance to the locations, the feasible area changes more which leads to the increase of costs. By increasing parameter λ , total profit decreases. This change interval leads to the decrease in total profit with a constant slope and the relation between this parameter and total profit is approximately linear. In fact, by increasing the control of each scenario, the ambulance is assigned in a responsive way in all scenarios and the costs decrease. On the other hand, by increasing parameter $\overline{d}_{is}$, total profit increases, and this changes are tangible and more

Table 10 The obtained results from solving instance problems

Instances	GAMS objective	SA objective	GAP (%)	CPU (seconds)	SA run time (seconds)
Small-sized	− 392.108	− 0403.526	2.91195283	16.263	5.56
Large-sized	5426.26	5384.10	0	1000	31.098

Table 11 Investigating the changes in parameters

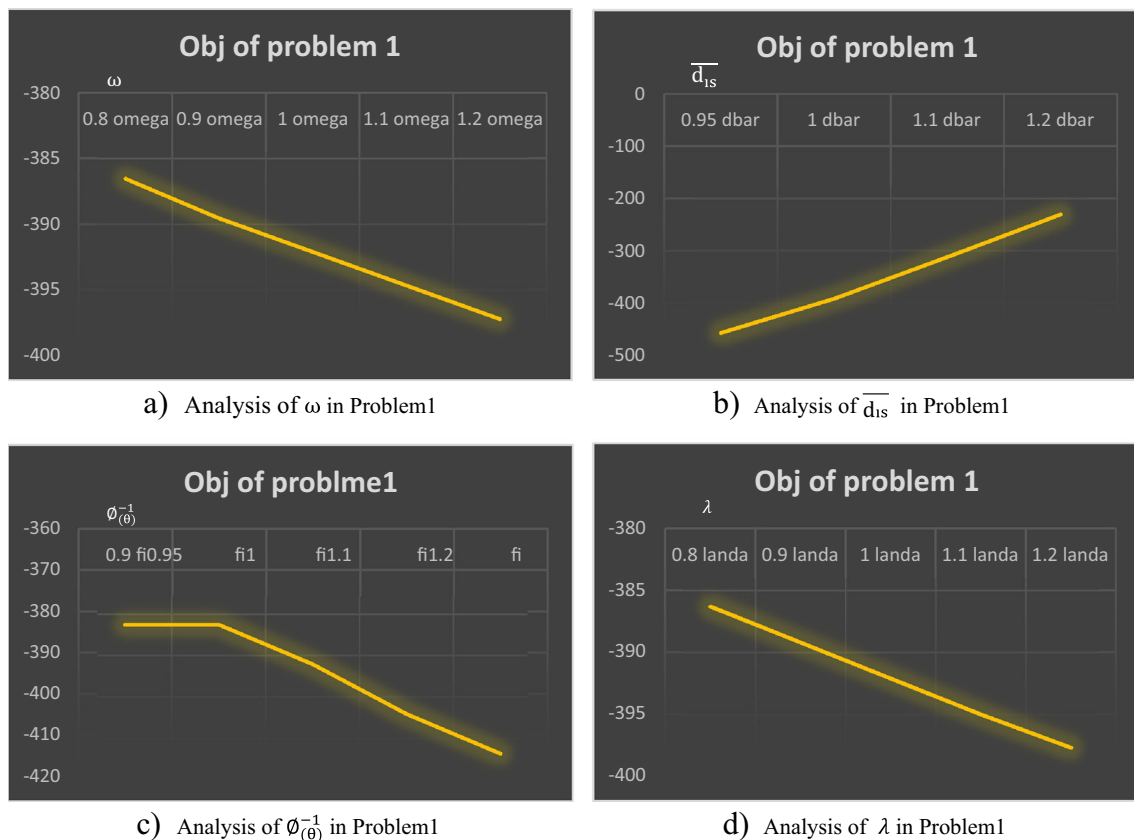
Objective function	Parameter changes intervals				
	− 20%	− 10%	0	+ 10%	+ 20%
Objective function in problem 1 for the changes of ω	− 386.536	− 389.546	− 392.108	− 394.670	− 397.232
Objective function in problem 1 for the changes of λ	− 386.290	− 389.199	− 392.108	− 395.017	− 397.702
Objective function in problem 1 for the changes of d_{is}	infeasible	infeasible	− 392.108	− 312.252	− 230.700
Objective function in problem 1 for the changes of $\phi_{(\theta)}^{-1}$	infeasible	− 382.500	− 392.108	− 404.164	− 413.772
Objective function in problem 2 for the changes of ω	− 511.448	− 512.400	− 513.352	− 514.304	− 515.256
Objective function in problem 2 for the changes of λ	− 514.658	− 514.005	− 513.352	− 512.699	− 512.046

sensitive in comparison with the other objective functions. The main cause of this is the decrease in demand satisfaction. By increasing parameter $\phi_{(\theta)}^{-1}$, total profit decreases and these changes are very tangible. By a 5% decrease, total profit increases and by a 10% decrease, it does not work. Also, by a 20% decrease, the problem becomes infeasible. The main reason for this is that the extent of assignments decreases based on the scenarios and as a consequence, controls of other scenarios decreases and the costs increase.

As it is clear in Fig. 10, by increasing parameter λ , the value of total objective increases and this change interval with constant slope leading to increase total objective in problem 2. By increasing the control on other scenarios, the costs

increases and the main reason is that assignments are not capable of minimizing the cost in each scenario. By increasing parameter ω , the value of the total objective decreases. This change interval leads to profit decreasing with an almost constant slope and the relation between this parameter and the total profit is approximately linear. But, the amount of changes here is lower than problem 1 and with increasing the assignment of ambulance to the emergency centers, the cost increases. On the other hand, because of the higher increase in this problem, the objective function gets lower value in comparison to the previous problem.

With regard to the results, the model of problem 1 is solved with the probable satisfaction of the demand which is the

**Fig. 9** Investigating the object function of the problem 1 against the changes of the feasible space

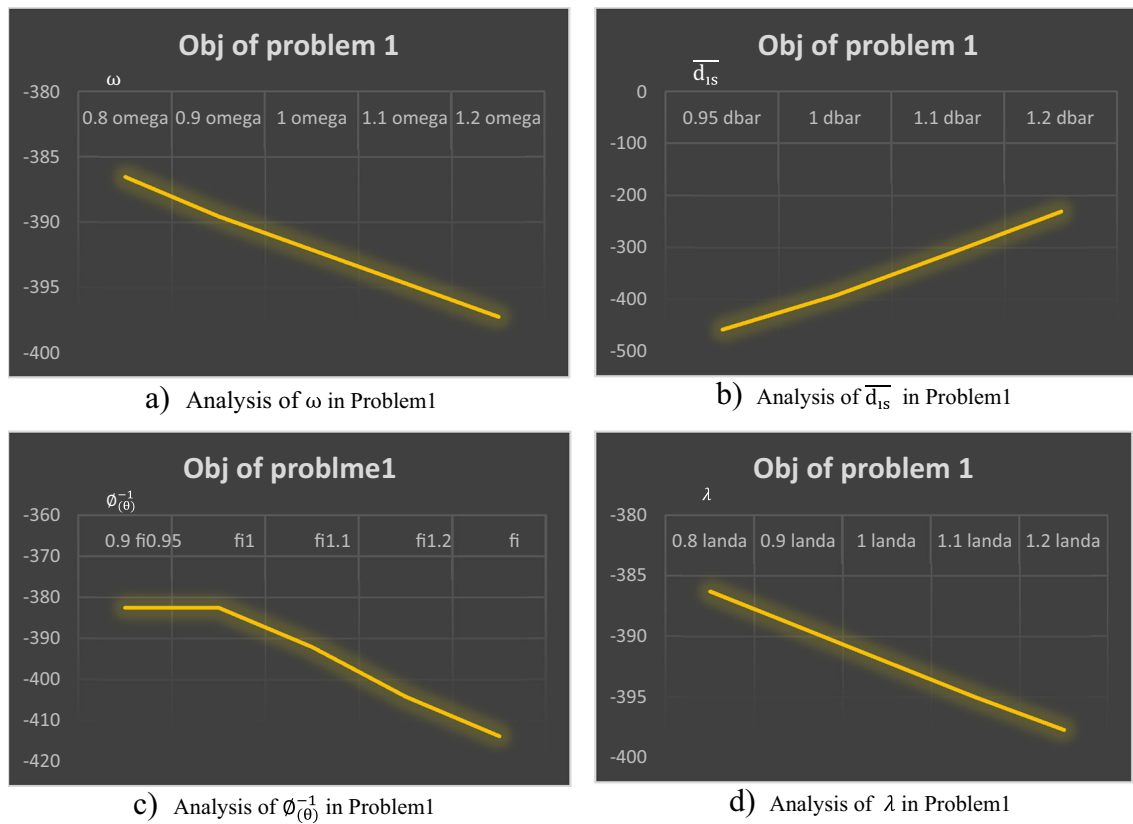


Fig. 10 Investigating the object function of the problem 2 against the changes of the feasible space

efficiency of the ambulance and the assigned ambulance can lead to control other scenarios and forms a stable system. On the other hand, in some figures, the slope increases or decreases and by analysis of these figures, it has been identified that which path is more appropriate if a decision should be made to decrease the costs. But, in fact, in some cases, it is impossible. Because an action can crave such a time and cost which to the decision maker is not able to afford. Therefore, declining slopes and increasing costs would be possible to be selected and indeed, the decision maker is obligated not to select the global optimal solution. Note that this problem has been studied upon this analysis and according to the environmental and execution limitation of the problem. In some cases, some parameters do not affect the final solution but they help a lot in problem orientation and control. Therefore, the main limitations of the problem can be known and the facilities can be utilized to achieve desirable results and prevent the waste of resources and time.

5. Conclusion

Relief has special importance in times of occurrence of incidents as they create risks to the lives. To decrease the risk and increase the survival chance of the injuries, the relief actions should take place at a standard time. On the other hand, to create an integrated relief system, there are limitations such as

financial resources. Also, to face the relief problem in a more logical way, the uncertainty of data should be considered in order to make the results closer to the reality. In this paper, the critical problem and relief logistics are investigated through eight steps. In the first step, with using system dynamics, the relief problem space is specified. In the second step, the causal space from control viewpoint is investigated. In the third step, by using simulation, demand of the ambulances is attained. In the fourth step, the probability of the occurrence of scenario events with regard to the game theory and according to the traffic and population density is calculated. In the fifth step, prioritization of choices for candidate emergency stations is done by TOPSIS and LINMAP model. In the sixth step, the integrated multi-step modeling of location of the emergency stations, allocation and routing of the ambulance according to fuzzy, and probability satisfaction of demand was studied. Also, considering uncertain resources for supplying the demand and fuzzy assignment costs with using the robust planning were investigated. In this step, the assignment of resources is according to the Poisson's distribution average which is obtained by estimating by means of a confidence interval. In the seventh step, a SA algorithm is developed in line with the defined problem. Finally, in eighth step, the problem results were analyzed using proposed solution method GAMS solver and SA algorithm. For future research, some suggestions are recommended as follows:

- Locating emergency centers in a dynamic manner considering time planning horizon
- Classifying ambulances with respect to their service types
- Considering supportive services for the ambulances. By applying a sensitivity analysis, the number of the required supportive ambulances is determined in each emergency location regarding the defined demand for increasing the confidence coefficient
- Determining the usage percentage of the ambulances, so that the emergency locations can lend their non-working ambulances to the locations with higher demand
- Adding maintenance of the ambulances and paths to the model
- For more equality, an objective function can be defined to decrease the uncovered demand

Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

Appendix 1. Control

According to Ogata's book, formulations are presented as below:

$$Mason \text{ interest} = \frac{\sum_i p_i \Delta_i}{\Delta} \quad (52)$$

Here, p_i is the forward path in signal diagram from input to the output, Δ_i represents the remaining path after traversing the forward path and Δ is the path with removing of the even loops and adding the odd loops. Therefore,

$$p_1 = (\alpha\%C^1)(1)\left(\frac{1}{D}\right)(c) \quad (53)$$

$$\Delta_1 = 1 \quad (54)$$

$$\Delta = 1 + \left[\alpha\%H(t) \frac{1}{D} \right] \quad (55)$$

$$Mason \text{ interest} = \frac{\alpha\%C^1 \times \frac{1}{D}}{1 + \alpha\% \frac{H(t)}{D}} \quad (56)$$

The transformation function is as follows:

$$\begin{aligned} G(s) &= \frac{+\alpha\% \frac{1}{s} C^1}{1 + \frac{1}{s}} = \frac{\alpha\%S^{-1}C^1}{1 + S^{-1}H(s)} \times S^2 \\ &= \frac{+\alpha\%SC^1}{S^2 + H(s)S} \end{aligned} \quad (57)$$

Considering the equations below,

$$\begin{aligned} \frac{1}{D} &= \frac{1}{S} \\ x(t) &= x(s) \\ H(t) &= H(s) \end{aligned} \quad (58)$$

The state equation is calculated as below:

$$\begin{aligned} a_1 &= H(s) \\ a_2 &= 0 \\ a_0 &= 1 \end{aligned} \quad (59)$$

$$\begin{aligned} b_0 &= 0 \\ b_1 &= \alpha\%C^1 \\ b_2 &= 0 \end{aligned} \quad (60)$$

$$\begin{aligned} \beta_0 &= 0 \\ \beta_1 &= b_1 = \alpha\%C^1 \\ \beta_2 &= b_2 - \alpha_1 \beta_1 = (0 - H(s)\alpha\%C^1 = -\alpha\%CHS) \end{aligned} \quad (61)$$

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -H(t) & 0 \end{bmatrix} x(t) + \begin{bmatrix} +\alpha\% \\ -\alpha\%H(s) \end{bmatrix} u \quad (62)$$

With regard to the stable state equation, the stability of the problem is investigated:

$$\text{row 1} \begin{bmatrix} a_0 & a_2 \\ 1 & 0 \end{bmatrix} \quad (63)$$

$$\text{row 2} \begin{bmatrix} a_1 & a_2 \\ H(s) & 0 \end{bmatrix} \quad (64)$$

$$\text{row 3} b_1 = 0 \quad (65)$$

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1} \quad (66)$$

$$b_1 = 0 - 0 = 0 \quad (67)$$

If decadence has been omitted unintentionally, the second solution would be as follows:

$$S^2 + H(s)S \Rightarrow \frac{1}{p} + \frac{Hs}{p} = 0 \Rightarrow p^2 + PH(s) = 0 \quad (68)$$

$$1 \text{ row } p \quad 0 \quad (69)$$

$$2 \text{ row } 1 \quad 0 \quad (70)$$

$$3 \text{ row } 0 \quad (71)$$

According the obtained results, the system is stable in both states.

Appendix 2. Determining the Probability of the Scenario Occurrence

For further explanation, formulations, and computations are presented as follows:

How to calculate P_i (i th probability): First, some definitions are described in Table 12.

Table 12 Definition of parameters and variables

x_i	Decision variable for player i
$f_i(x)$	Objective function for player i
$g_i(x)$	Constraint for player i
λ_i	Lagrangian coefficient
b_i	Source of player i
B_i	Right hand coefficient of player i
$R_i(x)$	Revenue
P_i	Occurrence probabilities for player i
e	Matrix 1
δ_i	Free variable
$x_j^i \in x_i, j = 1, 2$	Strategy j for player i
a_{ij}^i	Profit gained by a game for player i using strategy j from player i, and strategy j from another player i

Mathematical Programming Model for Variable Decision Maker i:

$$\text{Max } f(x)$$

$$\text{St } g_i(x) \geq 0$$

$$x_i \geq 0$$

Kuhn-Tucker (KKT) Conditions (I):

By applying KKT conditions, model can change as below:

$$\text{Min } \sum_i \lambda_i g_i(x)$$

$$\nabla f_i(x) + \sum \lambda_i g_i(x) = 0 \quad \forall i$$

$$g_i(x) \geq 0$$

$$\lambda_i \geq 0$$

Assume that:

$$g_i(x) : B_i x \leq b_i$$

$$\text{So, we have } \nabla g_i(x) = -B_i$$

On the other hand, we consider: $f_i(x) = R \times x = R_i(x)x$

Therefore, $\nabla f_i(x) = R_i(x)$

$$\text{Min } \sum_i \lambda_i (b_i - B_i x) \quad (\text{II})$$

$$\text{St. } R_i(x) - \sum \lambda_i B_i = 0 \quad (\text{III})$$

$$b_i - B_i(x) \geq 0$$

$$\lambda_i \geq 0$$

(IV)

With respect to (III), we have for (II):

$$\begin{aligned} R_i(x) &= \sum \lambda_i B_i \\ \rightarrow \underbrace{R_i(x)x_i}_{\sum_i f(x)} &= \sum_i \lambda_i B_i x_i \end{aligned}$$

After rewriting (IV):

$$\text{Min } \sum_i \lambda_i b_i - \sum_i f_i(x)$$

$$R_i(x) - \sum_i \lambda_i B_i = 0 \quad (\nabla)$$

$$b_i - B_i(x) \geq 0$$

$$\lambda_i \geq 0$$

$$\begin{array}{cc}
 & \text{Second Player} \\
 & x_1^2 \quad x_2^2 \\
 \text{First Player} & x_1^1 \begin{bmatrix} a_{11}^1 & a_{12}^1 \\ a_{21}^1 & a_{22}^1 \end{bmatrix}
 \end{array}$$

Analyze the problem with game condition:

x_1^1 : Strategy 1 from player 1
 a_{11}^1 : Profit gained by game for player 1 using strategy 1 from player 1, and strategy 1 from player 2

And objective for second player:

$$R_i(x) = (x_1^1 x_2^1) \begin{pmatrix} a_{11}^1 & a_{12}^1 \\ a_{21}^1 & a_{22}^1 \end{pmatrix}$$

$$\text{Desirabilities} : \begin{cases} a_{11}^1 x_1^1 + a_{21}^1 x_2^1 \\ a_{12}^1 x_1^1 + a_{22}^1 x_2^1 \end{cases}$$

By considering second player in above equation:

$$f_1(x) = (x_1^1 x_2^1) \cdot \begin{pmatrix} a_{11}^1 & a_{12}^1 \\ a_{21}^1 & a_{22}^1 \end{pmatrix} \cdot \begin{pmatrix} x_1^2 \\ x_2^2 \end{pmatrix}$$

$$f_2(x) = (x_1^1 x_2^1) \cdot \begin{pmatrix} a_{11}^2 & a_{12}^2 \\ a_{21}^2 & a_{22}^2 \end{pmatrix} \cdot \begin{pmatrix} x_1^2 \\ x_2^2 \end{pmatrix}$$

Then,

$$\sum_{i=1}^2 f_i(x) = (x_1^1 x_2^1) \left[\begin{pmatrix} a_{11}^1 & a_{12}^1 \\ a_{21}^1 & a_{22}^1 \end{pmatrix} + \begin{pmatrix} a_{11}^2 & a_{12}^2 \\ a_{21}^2 & a_{22}^2 \end{pmatrix} \cdot \begin{pmatrix} x_1^2 \\ x_2^2 \end{pmatrix} \right]$$

Now, we analyze randomness condition of game; according to occurrence probability for each player in above model:

$$\begin{aligned}
 & \left\{ \begin{array}{l} P_i \geq 0 \\ \sum P_i = 1, \text{ ep}_i = 1 \end{array} \right\} \begin{cases} \text{ep}_i \geq 1 \\ \text{ep}_i \leq 1 \end{cases} \\
 & \rightarrow \left\{ \begin{array}{l} (1) P_i \geq 0 \quad (S_i) \\ (2) \text{ep}_i - 1 \geq 0 \quad (\beta_i) \\ (3) 1 - \text{ep}_i \geq 0 \quad (\alpha_i) \end{array} \right\} \xrightarrow{\text{Lagrangian coefficients}} \text{according to KKT condition} \rightarrow b_i - B_i(x_i) \geq 0
 \end{aligned}$$

$$\begin{aligned}
 & \rightarrow b_i = \begin{pmatrix} 0 \\ +1 \\ -1 \end{pmatrix}, \quad B_i = \begin{pmatrix} -I \\ +e \\ -e \end{pmatrix}, \quad x_i = p_i \\
 & , e = (1, 1, \dots), \quad -e = (-1, \dots) \\
 & , \lambda_i \begin{pmatrix} s_i \\ \alpha_i \\ \beta_i \end{pmatrix}
 \end{aligned}$$

$$R_i(P_i) - \delta_i e \leq 0$$

$$P_i \geq 0$$

$$\text{ep}_i = 1$$

$$\delta_i = \text{free variable}$$

By replacing above formulations in equation (V2):

$$\text{Max} \sum E f_i(x) - \sum \delta_i$$

In the presented model, among all three sections of the city, the first and the third sections are competing with each other and their game tables are shown based on the two criteria of traffic and population density in

below. The second section, which is a transformation place, is concluded from the other two sections.

The first and second row (p_1 and p_2) are obtained from the first criterion:

$$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

For p_3 and p_4 which are obtained from the second criterion, we have

$$\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Thus, using the mathematical model, values for p are calculated as below (Wang et al. 2016):

$$\begin{aligned} & \text{Max } (p_1, p_2) \begin{bmatrix} 2+1 & -1-1 \\ -1-1 & 1+2 \end{bmatrix} \left(\frac{p_3}{p_4} \right) - (\rho_1 + \rho_2) \\ & \text{s.t.} \\ & (p_1, p_2) \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} - (\rho_1 + \rho_2) \leq 0 \\ & (p_3, p_4) \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} - (\rho_1 + \rho_2) \leq 0 \\ & p_1 + p_2 = 1 \\ & p_3 + p_4 = 1 \\ & p_1, p_2, p_3, p_4, \rho_1, \rho_2 \geq 0 \end{aligned} \quad (72)$$

where ρ_1 and ρ_2 are the Lagrange coefficients for each criteria and p_5 (which is shown in Table 5) is proportional to the sum of them.

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